# A Simple Method to Combine the Output Power From Multiple Class-E Power Amplifiers

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***Abstract*— Radio frequency (RF) power amplifiers are an integral part of many academic, medical, and industrial applications. For many of these applications, the required power level is high enough such that a single amplifier circuit cannot provide enough power, and multiple amplifiers are needed. Conventionally, a power combiner network is used to isolate any mismatch between the multiple power amplifiers, combining their output powers. However, these power combiner networks have their disadvantages, including additional loss and an increase in overall system size. In this article, we present an alternative approach to designing a high-efficiency high-power RF amplifier system. Here, we offer a simple tuning method that allows output powers from multiple power amplifiers to be combined directly without the need for an extra combiner network, given that the mismatches between the amplifiers are sufficiently small. The method is specifically developed for class-E power amplifiers at kilowatts power level and is demonstrated with a 1500-W, 40.68-MHz power amplifier design, utilizing six subcircuits directly combining power.**

***Index Terms*— Class-E amplifier, gallium nitride (GaN), power combining, radio frequency (RF) power amplifier (PA), RF generator.**

## I. Introduction

RADIO frequency (RF) power amplifiers (PAs) are employed in a multitude of applications ranging from RF communication, medical imaging, industrial plasma generation and processing, and so on. Traditionally, linear amplifiers, such as class-A, class-B, or class-AB, are used because of their design simplicity and linearity. However, as the need for higher efficiency arises, switched-mode designs, such as class-D [1]–[3], E [4], [5], F [6]–[8], E/F [9], and Φ [10]–[12], have risen in popularity. Unlike a linear PA, switched-mode design can reach very high efficiency but falls short with regards to input–output linearity [13].

Depending on the operating frequency and the amplifier design, a single PA will always be bounded in output power by various limiting processes that include thermal design, package parasitics, device current/voltage capability, and so on. When the power required in the system exceeds that a single PA circuit can generate, multiple PAs are needed [14], [15].

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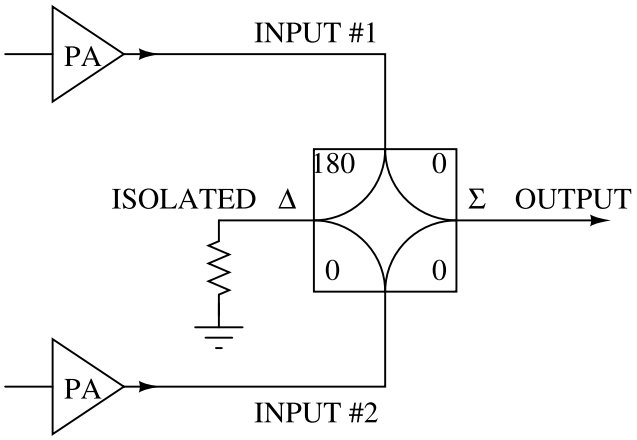



Fig. 1. Typical power combiner circuit consists of two input ports, one output port, and one isolated port.

To create a single output port, the output powers from these amplifiers have to be electrically combined. When the amplifiers are identical and driven with exactly the same inputs, it is possible to combine their output powers directly without the need for any added combiner network. At low enough frequency (<5 MHz), this is achievable with a good layout and a tight component variation control. However, at higher frequency (10+ MHz), any slight mismatch between the amplifiers, either due to variation in the component values or the input signal timing, can cause serious problems to direct connection.

To avoid problems that come from this mismatch, a separate power combiner network is typically employed to combine the output powers from two or more PAs. There are many ways to create a power combiner network. The most conventional approach is to use an isolating combiner [16]. In its most common form, an isolating power combiner network consists of two input ports, one output port, and one isolated port, as shown in Fig. 1. The matched portions of the input powers are combined and sent to the output port. The mismatched portions of the input powers are isolated out and sent to the isolated port. Conventionally, this is then dissipated away as heat on an isolation resistor; however, it can also be recovered back through a rectifier to improve the overall system efficiency [17]–[19].

Another approach to power combining is to use a lossless nonisolating combiner network where the isolation port is not needed. The most well-known circuit for this type of combiner

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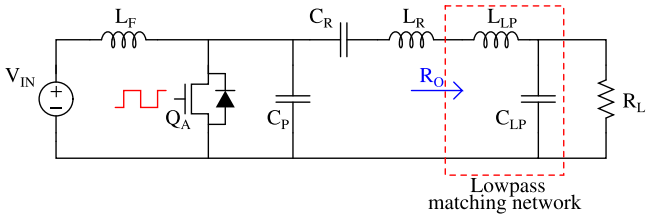



Fig. 2. Typical class-E PA circuit for high-power applications utilizing a low-pass matching network at the output.

is the classic Chireix combiner [16], [20], [21], but other nonisolating combiner typologies with improved performance have also been proposed [13], [22]–[24].

While using power combiner networks is a good solution to the mismatch problem when one needs to combine the power from multiple amplifiers, they do present their own disadvantages. Regardless of the type, a combiner network consists of inductors, capacitors, or transmission lines. Under operation, each of these components will present their associating loss due to their parasitic resistances. Thus, even with perfectly matched inputs, a "lossless" combiner network is never truly lossless, causing the efficiency at the combined output to always be slightly lower than the original efficiency of each PA. Furthermore, for applications where space is limited or where weight is critical (e.g., RF communication for cell phones and microsatellite propulsion), these components do take up extra valuable space and add excess weight to the system. For certain types of combiner networks, the number of combinable PAs must also follow a binary sequence ($2^N = 2, 4, 8, 16, \ldots$), limiting the design freedom.

In this work, we present an alternative approach to the problem of power combining. Here, we offer a simple methodology to tune and operate individual PA stages that allow their output powers to be combined directly without the need for additional power combiner networks. This method eliminates the extra loss associated with the combiner network, reduces the system size and weight, and allows the circuit designer freedom to connect any number of PAs together, given that their mismatch is sufficiently small.

Section II mathematically derives the necessary condition for direct power combining in the proposed tuning method, as well as describes the steps to convert a standard class-E amplifier into a "power-combinable" class-E amplifier. Utilizing this tuning technique, Section III demonstrates a continuous wave (CW) 1500-W, 40.68-MHz PA design with six class-E amplifiers directly combining power. The system achieves a high dc-to-RF efficiency of 89% and is intended for plasma generation applications. While the equations and method developed in this article mainly target class-E PAs used in high-power (kilowatts level) applications, they can be easily modified to be used with other types of amplifiers at other power levels. Finally, Section IV concludes this article.

## II. Power Combining Without a Power Combiner

Fig. 2 shows the schematic of a typical class-E PA circuit for high-power applications. Because a class-E amplifier appears as a voltage source, in order to output high power, the output resistance that the amplifier sees, $R_O$, needs to be of a small value. However, in RF applications, the actual load of the circuit, $R_L$, is commonly a 50-Ω impedance [25]–[27].

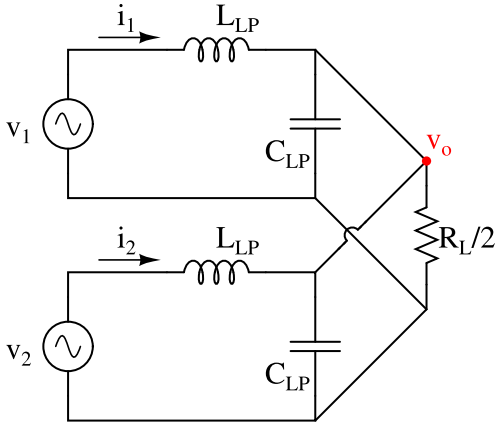



Fig. 3. Simplified circuit representing two amplifiers with *LC* low-pass networks combining power at the output.

As a result, a low-to-high matching network is usually needed to match these two impedances. In this article, we will demonstrate the case where a low-pass *LC* matching network is utilized; however, the derived equations can be easily modified for when other types of matching networks are used.

For a low-pass low-to-high *LC* matching network, the following textbook equations calculate the capacitor and inductor value needed for proper impedance matching [28]:

$$C_{\mathrm{LP}} = \frac{1}{\omega R_{\mathrm{L}}}\sqrt{\frac{R_{\mathrm{L}}}{R_{\mathrm{O}}} - 1} \tag{1}$$

$$L_{\mathrm{LP}} = \frac{R_{\mathrm{O}}}{\omega}\sqrt{\frac{R_{\mathrm{L}}}{R_{\mathrm{O}}} - 1}. \tag{2}$$

Instead of tuning the matching network to simply match the load to the optimum impedance that the PA can drive as per (1) and (2), here, we present an alternative way to pick the inductor and capacitor values. On top of performing impedance matching, this alternative network makes direct power combining possible even when there is a timing mismatch between the PAs. Sections II-A and II-B will explain the derivation of this modified low-pass matching network and how it can be used in a class-E amplifier circuit.

### A. Modified Low-Pass Matching Network

In this section, we will analyze the case with two amplifier circuits combining power; however, the result can be directly applied to any number (2+) of circuits combining power. Fig. 3 shows the simplified circuit, with $v_o$ being the output voltage across the combined load, $R_L/2$. To simplify the problem, we will approximate the output of each amplifier as a sinusoidal voltage source with the same amplitude, $V$. Furthermore, we will assume that due to a small timing mismatch between the gate driving signals of the two class-E circuits,

the two voltage sources have a small phase difference $\delta$

$$v_1 = Ve^{j(\omega t+\phi_1)}$$
$$v_2 = Ve^{j(\omega t+\phi_2)}$$
$$\delta = \phi_2 - \phi_1 \ll 1.$$

To minimize the effect of this timing mismatch on the amplifiers, we want to find the $L_{\rm LP}$ and $C_{\rm LP}$ values that allow the output power from each voltage source to remain the same regardless of the change in $\delta$.

To find the output power from the top voltage source, $P_1$, we first find the output voltage, $v_{\rm o}$, by using super-position

$$v_{\rm o} = Ve^{j\omega t}(e^{j\phi_1}+e^{j\phi_2})\frac{R_{\rm L}/2}{\left(-\omega^2 R_{\rm L}L_{\rm LP}C_{\rm LP}+R_{\rm L}+j\omega L_{\rm LP}\right)}. \tag{3}$$

Then, we can calculate the top source current $i_1$ from

$$i_1 = \frac{v_1 - v_{\rm o}}{j\omega L_{\rm LP}}.$$

Substituting $v_{\rm o}$ from (3) gives

$$i_1 = \frac{Ve^{j(\omega t+\phi_1)}}{j\omega L_{\rm LP}}\left(1-\frac{(1+e^{j(\phi_2-\phi_1)})R_{\rm L}/2}{(-\omega^2\ R_{\rm L}L_{\rm LP}C_{\rm LP}+R_{\rm L}+j\omega L_{\rm LP})}\right).$$

The output power from the top source, $P_1$, can be calculated from

$$P_1 = \mathbf{Re}\left[\frac{v_1\ i_1^*}{2}\right].$$

Hence

$$P_1 = \mathbf{Re}\left[-\frac{V^2}{2j\omega L_{\rm LP}}\left(1-\frac{(1+e^{-j(\phi_2-\phi_1)})R_{\rm L}/2}{(-\omega^2 R_{\rm L}L_{\rm LP}C_{\rm LP}+R_{\rm L}-j\omega L_{\rm LP})}\right)\right].$$

By using a small phase difference approximation, we get

$$P_1 = \mathbf{Re}\left[-\frac{V^2}{2j\omega L_{\rm LP}}\left(1-\frac{(2-j\delta)R_{\rm L}/2}{(-\omega^2 R_{\rm L}L_{\rm LP}C_{\rm LP}+R_{\rm L}-j\omega L_{\rm LP})}\right)\right].$$

Thus

$$P_1 = \frac{V^2}{2}\frac{\left(R_{\rm L}+\dfrac{\delta R_{\rm L}^2\omega C_{\rm LP}}{2}-\dfrac{\delta R_{\rm L}^2}{2\omega L_{\rm LP}}\right)}{(\omega L_{\rm LP})^2+(-\omega^2 R_{\rm L}L_{\rm LP}C_{\rm LP}+R_{\rm L})^2}. \tag{4}$$

To find the condition where $P_1$ does not change in relative to $\delta$, we set

$$\frac{d}{d\delta}P_1 = 0.$$

As a result, we find that

$$\omega^2 L_{\rm LP}C_{\rm LP} = 1 \tag{5}$$

and

$$P_1 = \frac{V^2}{2}\frac{R_{\rm L}}{(\omega L_{\rm LP})^2}. \tag{6}$$

By symmetry, the same also applies to the output power from the bottom source, $P_2$.

Equation (5) means that if we pick $C_{\rm LP}$ and $L_{\rm LP}$ such that they resonate at the switching frequency, then they will "shield" the circuits from the small timing mismatch between PAs, enabling output power from multiple PAs to be directly combined. Furthermore, (6) also dictates that when $C_{\rm LP}$ and $L_{\rm LP}$ are tuned this way, the output power will be linearly proportional to the load resistance. Essentially, this tuning makes the voltage source appears to the load as a current source.

While this condition on $C_{\rm LP}$ and $L_{\rm LP}$ enables direct power combining, there are infinitely many possible combinations of them. To maintain the impedance matching functionality that the original low-pass matching circuit provides, we have to find the values of $C_{\rm LP}$ and $L_{\rm LP}$ such that the output powers from the two voltage sources $v_1$ and $v_2$ are equal to the original output powers from the conventional matching network in (1) and (2).

In the original case, the impedance looking into the low-pass matching network is $R_{\rm O}$. Thus, the original output power from each source is

$$P_{1,\rm ori} = P_{2,\rm ori} = \frac{V^2}{2}\frac{1}{R_{\rm O}}.$$

By setting $P_{1,\rm ori} = P_1$ from (6), we find that, in order to maintain the same power, the modified low-pass network must have

$$C_{\rm LP} = \frac{1}{\omega\sqrt{R_{\rm L}R_{\rm O}}} \tag{7}$$

$$L_{\rm LP} = \frac{\sqrt{R_{\rm L}R_{\rm O}}}{\omega}. \tag{8}$$

Naturally, this choice of $C_{\rm LP}$ and $L_{\rm LP}$ does not guarantee that the input impedance looking into this modified low-pass network will be resistive. In fact, the impedance looking into this network is slightly inductive

$$Z_{\rm IN} = \frac{R_{\rm L}}{\left(1+\omega^2 C_{\rm LP}^2 R_{\rm L}^2\right)}+\frac{j\omega\left(L_{\rm LP}\left(1+\omega^2 C_{\rm LP}^2 R_{\rm L}^2\right)-C_{\rm LP}R_{\rm L}^2\right)}{1+\omega^2 C_{\rm LP}^2 R_{\rm L}^2}$$
$$Z_{\rm IN} = \frac{R_{\rm O}}{1+\frac{R_{\rm O}}{R_{\rm L}}}\left(1+j\sqrt{\frac{R_{\rm O}}{R_{\rm L}}}\right). \tag{9}$$

To maintain zero-voltage-switching (ZVS) condition in a class-E amplifier, an extra capacitance should then be added to the $C_{\rm P}$ to counteract this inductive component.

### *B. Power-Combinable Class-E Amplifier*

In this section, we will describe the tuning steps to create power-combinable class-E amplifiers by utilizing the derived modified matching network. As an example, Fig. 4 shows a system with four power-combinable class-E amplifiers directly combining their power at the output.

The tuning of power-combinable class-E amplifiers starts by selecting the component values, $L_{\rm F}$, $C_{\rm P}$, $L_{\rm R}$, and $C_{\rm R}$, as well as the output impedance, $R_{\rm O}$, following standard class-E design equations according to the desired operation regime and power level. For example, according to Acar *et al.*, if one wishes to attain the maximum output power from a given switching device, one should use the following component

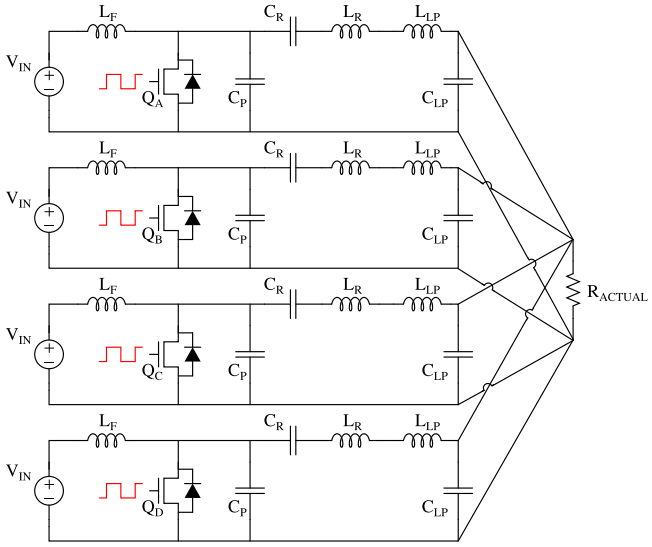



Fig. 4. System consists of four class-E amplifiers combining power at the output.

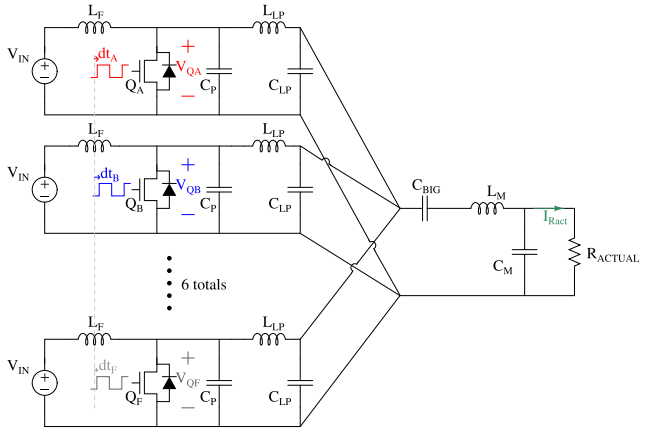



Fig. 5. Six class-E amplifier circuits with double-stage low-pass matching networks combining power.

values:

$$R_{\rm O} = \frac{k_{\rm P} V_{\rm in}^2}{P}, \quad C_{\rm P} = \frac{k_C}{\omega R_{\rm O}}, \quad L_{\rm F} = \frac{k_{\rm L} R_{\rm O}}{\omega}$$
$$L_R = \frac{Q R_{\rm O}}{\omega}, \quad C_R = \frac{1}{\omega Q R_{\rm O}}$$
$$(k_{\rm P} = 1.365,\ k_C = 0.685,\ k_{\rm L} = 0.732) \quad (10)$$

where $Q$ is the desired loaded quality factor of the series output filter [29].

Depending on the required actual load impedance $R_{\rm actual}$ and the number of circuits $N$ to be combined, the load impedance each circuit has to be tuned for, $R_{\rm L}$, can be calculated from

$$R_{\rm L} = N \times R_{\rm actual}. \quad (11)$$

Using both the calculated $R_{\rm O}$ and $R_{\rm L}$, we can calculate the modified low-pass component values from (7) and (8)

$$C_{\rm LP} = \frac{1}{\omega\sqrt{R_{\rm L} R_{\rm O}}}, \quad L_{\rm LP} = \frac{\sqrt{R_{\rm L} R_{\rm O}}}{\omega}.$$

As previously stated, in order to compensate for the added inductive loading due to the modified low-pass network, an extra capacitance $C_{\rm extra}$ should be added to the $C_{\rm P}$ such that ZVS condition is attained. The value of this $C_{\rm extra}$ depends on the type and operation regime of the selected class-E amplifier. However, because ZVS condition in a class-E circuit depends not only on its fundamental contents but also on the exact shape of the drain voltage and current waveforms, the exact value of $C_{\rm extra}$ is best obtained via circuit simulation. Nonetheless, we have empirically found that for a class-E amplifier tuned according to the (10), the following $C_{\rm extra}$ value will result in a very near ZVS condition:

$$C_{\rm extra} \approx C_{\rm P}\sqrt{\frac{R_{\rm O}}{R_{\rm L}}}. \quad (12)$$

Thus, the total value of the required parallel capacitance for each maximum output power class-E circuit becomes

$$C_{P,\rm tot.} \approx \frac{k_C}{\omega R_{\rm O}}\left(1 + \sqrt{\frac{R_{\rm O}}{R_{\rm L}}}\right). \quad (13)$$

## III. Experimental Demonstration

In this section, we will experimentally demonstrate the design of a 40.68-MHz 1500-W RF amplifier utilizing the proposed power-combinable class-E amplifier circuit. The intended application for this RF amplifier is for the plasma generation used in semiconductor processing industries for etching and deposition [30]. In order to reach the desired 1500-W power level while keeping the circuit temperature sufficiently low, we choose to use six identical class-E circuits combining the power to spread out the heat generation due to the loss on the switches. To get the best performance at this high frequency, 650-V gallium nitride (GaN) transistors, GS66506T(s), from GaNSystems are used as the main power switches.

### A. Circuit Design and Simulation

Fig. 5 shows the schematic of the circuit used in this prototype. Here, the terms $dt_X$(s) represent the timing mismatch/delay between the gating signals of the different amplifiers. There are many causes of this timing mismatch, with the variation of the propagation delays in the logic gate and gate driver chips being the most common source. Unlike in Section II, here, we choose to use two-stage low-pass matching networks in order to keep the quality factors of the matching networks low, making them less sensitive to component variation in an actual circuit. The first stage ($L_{\rm LP}$ and $C_{\rm LP}$) is tuned according to the proposed modified low-pass network, while the second stage ($L_{\rm M}$ and $C_{\rm M}$) is tuned as a conventional matching network. Furthermore, unlike in Section II where series $LC$ filters are used to filter out the other harmonic contents, a single large capacitor $C_{\rm BIG}$ is used here for dc-blocking. Since the two-stage low-pass network provides adequate filtering, the removal of this series resonant

TABLE I
OPERATING CONDITION AND COMPONENT VALUES OF THE SIMULATED POWER-COMBINABLE CLASS-E AMPLIFIER CIRCUITS IN FIG. 5

| $V_{IN}$ [V] | FET | $L_F$ [nH] | $C_P$ [pF] | $L_{LP}$ [nH] | $C_{LP}$ [pF] | $L_M$ [nH] | $C_M$ [pF] | $C_{BIG}$ [nF] | $R_{act.}$ [Ω] |
|---|---|---|---|---|---|---|---|---|---|
| 60 | GS66506T | 42 | 150 | 130 | 115 | 85 | 135 | 4 | 50 |

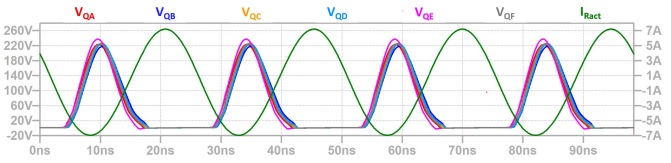



Fig. 6. Drain waveforms (red/blue/orange/sky blue/magenta/gray) and output currents (green) of the simulated power-combinable class-E amplifier circuits.

filter serves to further increase the efficiency and reduce the overall size/volume of the circuit.

The general design step follows that what is outlined in Section II. However, to take into account losses in the switches and passive components as well as the nonlinearity of the junction capacitance of the transistor, we overdesign the circuit by tuning it with the equation for power level $P = 2000$ W instead of 1500 W. This results in the $L_F$ and $C_P$ values that are roughly 30% lower/higher, as well as the $L_{LP}$ and $C_{LP}$ values that are roughly 15% lower/higher than those of an ideal circuit.

We pick the input voltage of the circuit to optimize the circuit efficiency in the face of "OFF-state" $C_{OSS}$ loss in GaN devices according to [31]–[33]. With 60-V input voltage and 2000-W power, we can calculate from (10)

$$R_O = \frac{k_P V_{in}^2}{P} = \frac{1.365 \cdot 60^2}{2000/6} = 14.7\ \Omega$$
$$L_F = \frac{k_L R_O}{\omega} = \frac{0.732 \cdot 14.7}{2\pi \cdot 40.68 \cdot 10^6} = 42.3\ \text{nH}.$$

In order to keep a similar matching ratio (quality factor) of both matching stages, the first stage matching network is tuned to match the 14.7-Ω output impedance of the amplifier ($R_O$) to a 75-Ω impedance ($R_L$). Then, the second stage matching network is tuned to match the combined impedance of six 75 Ω in parallel (75/6 = 12.5 Ω) to the actual load impedance of 50 Ω.

Following (7), (8), and (13), we find

$$C_{LP} = \frac{1}{2\pi \cdot 40.68 \cdot 10^6 \sqrt{75 \cdot 14.7}} = 118\ \text{pF}$$
$$L_{LP} = \frac{\sqrt{75 \cdot 14.7}}{2\pi \cdot 40.68 \cdot 10^6} = 130\ \text{nH}$$
$$C_{P,\text{tot.}} = \frac{0.685}{2\pi \cdot 40.68 \cdot 10^6 \cdot 14.7}\left(1+\sqrt{\frac{14.7}{75}}\right) = 262\ \text{pF}.$$

Then, minor adjustment is then made on simulation to achieve the best performance and to take into account the junction capacitance of the switches. Table I summarizes the component values used in the simulation. For this simulation, a quality factor of 150 is used for all inductors, while a quality factor of 1000 is used for all capacitors.

To achieve a realistic simulation, we randomly insert a 5% variation in the component values into different passive components. Variation of the gate timing, $dt_A - dt_F$, of up to 1 ns is also randomly added to the different gate signals to simulate timing mismatch. From our experience and in consistent with device datasheets [34], [35], this number is a good approximation for typical variation in gate timings due to the propagation delays in the discrete ICs unless intensive component matching is performed.

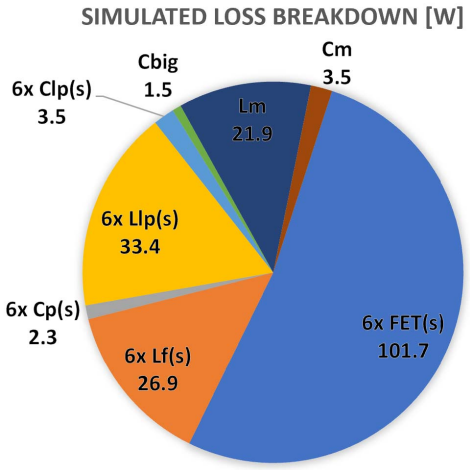



Fig. 7. Simulated loss breakdown of the amplifier shown in Fig. 5.

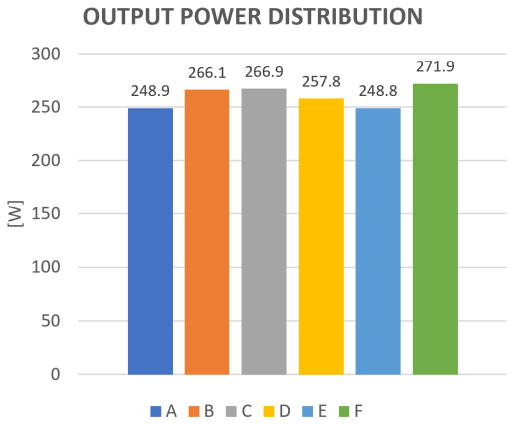



Fig. 8. Simulated output power distribution of the amplifier shown in Fig. 5.

Fig. 6 shows the simulated drain waveforms across the six switches, as well as the sinusoidal output current across the 50-Ω load. Notice that due to the introduced component value and timing variation, the drain waveforms from each of the six subcircuits slightly differ. Fig. 7 shows the simulated loss breakdown across the different components. The simulated output power is 1533 W, and the simulated drain efficiency is 88.5%. Figs. 8 and 9 show the simulated output power distribution and transistor loss distribution across the six circuits (A to F). Even with both the component value variation and the timing mismatch, the output power and the transistor

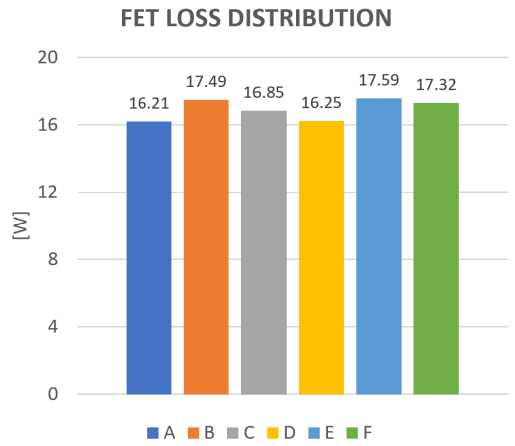
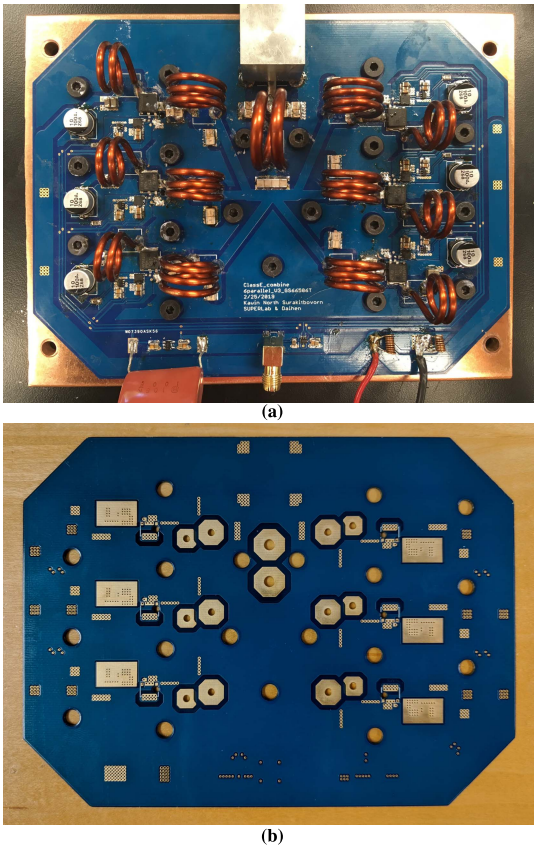



Fig. 9. Simulated transistor loss distribution of the amplifier shown in Fig. 5.



(a)

(b)

Fig. 10. 1500-W, 40.68-MHz amplifier circuit. The board dimension is 160×120 mm. (a) Front side (with components). (b) Back side (bare board).

loss values are still maintained to within less than 10% of each other.

### *B. Experimental Result*

The actual circuit shown in Fig. 10 is implemented on a standard two-layer FR-4 printed circuit board (PCB) using the component values from the simulation. All six subcircuits share the same gating signal, with six noninverting logic gate IC(s) used to buffer the signals. All the inductors are air-cored and hand-wound. TEMCo 10 AWG copper magnet wires are used for $L_F$(s) and $L_{\mathrm{LP}}$(s), and TEMCo 8 AWG copper magnet wire is used for the $L_{\mathrm{M}}$. Low-loss C0G ceramic capacitor from American Technical Ceramic (ATC) is used for $C_P$(s) $C_{\mathrm{LP}}$(s) and $C_{\mathrm{M}}$. Due to the high operating frequency, parasitic inductance and capacitance of the PCB traces also affect the tuning and are taken into account.

To sufficiently drive the gate of the tested GaN HEMTs, a low-side gate driver, LM5114 from Texas Instruments, is used. We place the gate drivers as close as possible to the GaN switches to minimize the loop inductance in the gate, as well as use a gate resistor of 2.4 Ω to dampen any ringing in the gate voltage. In addition, SiC Schottky diodes STPSC406 from ST Microelectronics, seen on the top of the board, are added in parallel to the switches to reduce the loss associated with reverse conduction in GaN transistor [36].

To facilitate the cooling of the switching devices, we mount the GaN transistors, GS66506T from GaNSystems, on the bottom side of the board so that they can be directly connected to the copper heat spreader. In addition, some inductors are also mounted as "through-holes" so that they can be in direct thermal contact with the thermal pad underneath the PCB. Water cooling, which is standard for commercial RF systems at this power level, is also used.

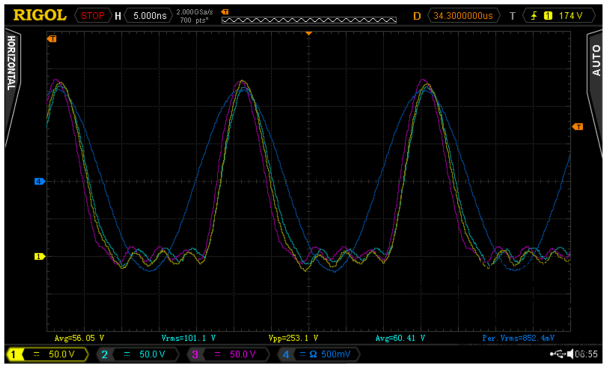
The input power is calculated from the dc input voltage and current under continuous operation. The input voltage is measured with a digital multimeter (DMM), Agilent 34411A, and the input current is recorded from the readout of the dc power supply, Agilent N5767A. The output power into the 50-Ω load is measured with a directional coupler/power meter setup. The setup consists of a calibrated four-port RF directional coupler, C5827-10, from Werlatone Inc., two N8482A thermocouple power sensors from Keysight Technologies, and an N1914A EPM series power meter from Keysight Technologies.

Fig. 11 shows the measured drain waveforms on all six subcircuits, as well as the output voltage waveform across an attenuator. While all the drain voltages show the characteristic soft-switching class-E waveforms, they differ slightly from each other due to the intrinsic mismatch/variation in the system. Nonetheless, with the proposed modified matching network, the circuit is able to attain a high dc-to-RF efficiency of 89±2% with 1560-W output power.

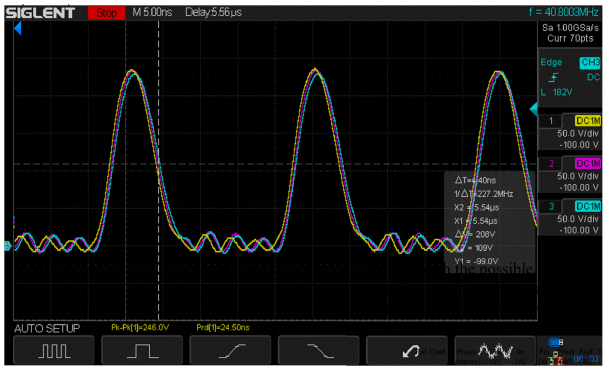
Fig. 12 shows the surface temperature of the passive components on top of the board under the steady-state operation. From the thermal image, the inductors of the units near the center of the board appear hotter than those on the outside. We believe that this is due to the natural temperature gradient on the PCB, with the center area being hotter than the area on the outer edges, and not from uneven power distribution between units. To measure the temperature of the switches that are mounted on the bottom side of the board, thermocouples are be inserted through small drill holes on the PCB. This thermal measurement shows all six FETs having their case temperature within 40 ± 3 °C, suggesting balanced power sharing between the six circuits.

### *C. Power-Sharing Verification*

To verify that the power is indeed equally shared amount the six stages, we perform an additional experiment where the dc




(a)




(b)

Fig. 11. Experimentally measured waveforms—two separate oscilloscopes are used to capture all of the waveforms. (a) Measured drain waveforms on units A, B, and C (yellow, sky blue, and pink) on the left-hand side of the board, as well as the output voltage (blue—measured across an attenuator). (b) Measured drain waveforms on units D, E, and F (yellow, pink, and sky blue) on the right-hand side of the board.

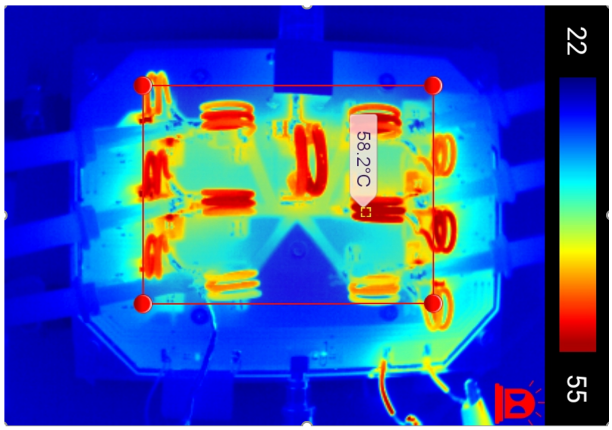



Fig. 12. Thermal image of the top side of the circuit under steady-state operation at full power.

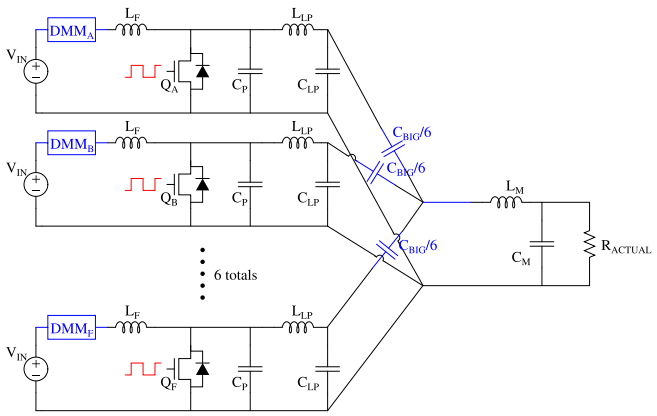



Fig. 13. Schematic of the circuit modified for individual dc-input power measurement.

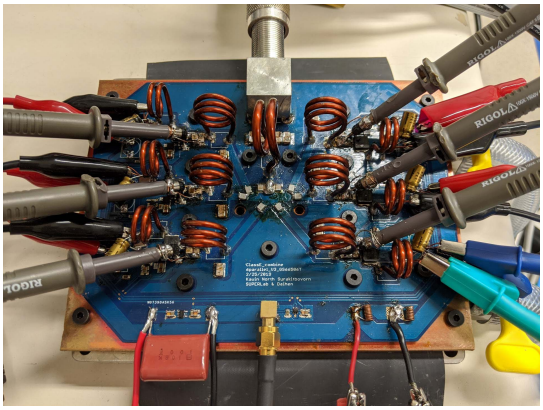


Fig. 14. Modified board used for the power-sharing verification experiment.

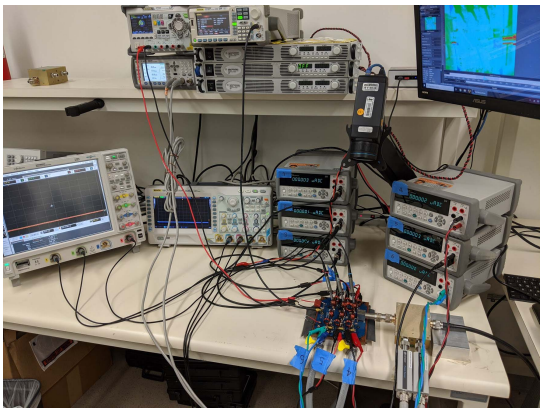


Fig. 15. Experimental setup for power-sharing verification.

input power into each unit is measured. Small modifications to the circuit are made to allow for this measurement without changing the operation of the original circuit. Especially, the dc-blocking capacitor $C_{BIG}$ is split into each branch of the circuit to dc-isolate the units from each other. Furthermore, the dc input trace on each unit is also modified so that a DMM can be inserted inline to measure the input current into each unit.

Fig. 13 shows the schematics of the modified circuit, while Fig. 14 shows the physical modification made to the PCB itself. As shown in Fig. 15, six DMMs, Agilent 34411A, are used to measure the input current into each unit. The dc input power into each unit is calculated from this current multiplied by the dc input voltage taken from the readout of the dc power supply; 12 data points are collected as we raise the input voltage from 5 to 60 V to increase the output power. The plot in Fig. 16 shows how the input power percentage (input power

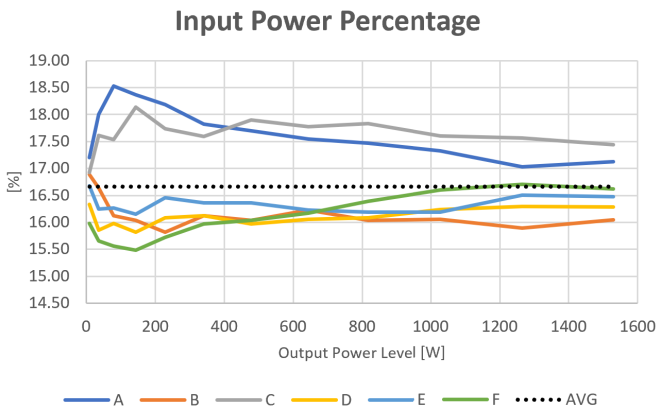



Fig. 16. Experimentally measured input power percentage of each unit ($y$-axis) versus output power level ($x$-axis).

into each unit divided by the total input power) changes as the power level is varied. The power-sharing capability appears to become more even as the power level increases. Once again, the highest and lowest measured input power percentages at full power fall within 10% of each other, confirming the balanced power sharing between the six circuits.

## IV. Conclusion

This article describes a new tuning method that allows output power from multiple amplifiers to be directly connected without the need for any additional power combiner networks. In contrast with conventional high-power RF amplifier systems where separate power combiner circuits are used to combine the output powers, the method introduced here relies on changing the way the components already existed in the high PAs are tuned without adding any extra components.

The major benefits of this tuning method are the simplicity of the design and the reduction in the number of components needed to achieve power combining capability. This results in circuits that are easy to design and scale, small in size and low in component counts, and can achieve much higher efficiency than conventional amplifiers.

The application of this modified matching network is demonstrated on class-E PAs to create power-combinable class-E amplifiers. The circuit is tested under continuous operation and shows a stable operation both thermally and electrically. In particular, we demonstrated a 1500-W RF amplifier with 89% efficiency and 25-W/in$^3$ power density at 40.68 MHz utilizing six class-E PAs combining power. This achieved efficiency is much higher than other studies at similar frequency and power level (79%–100 W–27.12 MHz [37], 83%–490 W–27.12 MHz [38], and 83%–1150 W–40.68 MHz [39]). Moreover, the achieved power density is also more than three times higher than the most power-dense RF generator commercially available, APEX RF Generator (7.4 W/in$^3$–5500 W–13.56 MHz [40]). While this article only provides explicit design equations for the low-pass matching network and class-E PA combination, the methodology derived here can be directly applied to other types of matching networks and other classes of amplifiers.

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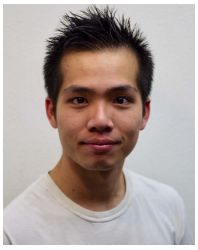
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